

# IBM SPSS Statistics 28.0 Output Report

Study: Neuro-Cognitive Dynamics of AI Attribution in Music Perception

Investigator: Berke Tan Tabak (IUE Psychology 2026)

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## T-Test: Paired Samples Statistics

| Pair Variable (N=30)           | Mean | N  | Std. Dev. | Std. Error Mean |
|--------------------------------|------|----|-----------|-----------------|
| Pair 1: Authenticity (Human)   | 5.67 | 30 | 0.78      | 0.14            |
| Pair 1: Authenticity (AI)      | 4.86 | 30 | 0.86      | 0.16            |
| Pair 2: Emotional Inv. (Human) | 4.99 | 30 | 1.11      | 0.20            |
| Pair 2: Emotional Inv. (AI)    | 3.97 | 30 | 0.94      | 0.17            |
| Pair 3: Quality (Human)        | 5.15 | 30 | 0.69      | 0.13            |
| Pair 3: Quality (AI)           | 4.62 | 30 | 0.76      | 0.14            |
| Pair 4: Liking (Human)         | 4.63 | 30 | 0.94      | 0.17            |
| Pair 4: Liking (AI)            | 4.68 | 30 | 0.95      | 0.17            |

## Paired Samples Test (Inferential Statistics)

| Paired Differences (Human - AI) | Mean Diff. | Std. Dev. | t      | df | Sig. (2-tailed) |
|---------------------------------|------------|-----------|--------|----|-----------------|
| Pair 1: Authenticity            | 0.806      | 0.348     | 12.692 | 29 | < .001          |
| Pair 2: Emotional Inv.          | 1.022      | 0.395     | 14.150 | 29 | < .001          |
| Pair 3: Quality                 | 0.533      | 0.346     | 8.449  | 29 | < .001          |
| Pair 4: Liking                  | -0.044     | 0.371     | -0.656 | 29 | .517            |

## Repeated Measures ANOVA: Tests of Within-Subjects Effects

Dependent Variable: Trust in Producer Label

| Source of Variation | Type III Sum of Sq. | df | Mean Square | F      | Sig.   | Partial Eta Sq. |
|---------------------|---------------------|----|-------------|--------|--------|-----------------|
| Label               | 0.022               | 1  | 0.022       | 0.041  | .841   | .001            |
| Error (Label)       | 15.728              | 29 | 0.542       |        |        |                 |
| Genre               | 12.441              | 2  | 6.221       | 11.850 | < .001 | .290            |
| Error (Genre)       | 30.426              | 58 | 0.525       |        |        |                 |
| Label * Genre       | 51.850              | 2  | 25.925      | 14.862 | < .001 | .339            |
| Error (Label*Genre) | 101.150             | 58 | 1.744       |        |        |                 |

# Linear Mixed Model (LME) Estimates: Right PFC Peak HbO

Dependent Variable: Right\_PFC\_Peak\_HbO (valid epochs, n=13 participants, N=149 trials)

## Model Dimension

| Parameter Type                  | Number of Levels | Covariance Structure          |
|---------------------------------|------------------|-------------------------------|
| Fixed Effects: Intercept, Label | 2                |                               |
| Random Effects: Participant     | 13               | Identity (Variance Component) |

## Estimates of Fixed Effects

| Parameter  | Estimate | Std. Error | df    | t     | Sig.              |
|------------|----------|------------|-------|-------|-------------------|
| Intercept  | 0.1815   | 0.0215     | 12.4  | 8.441 | < .001            |
| Label = AI | 0.0547   | 0.0270     | 135.2 | 2.026 | .047 (one-tailed) |

## Estimates of Covariance Parameters (Random Effects)

| Parameter            | Estimate | Std. Error | Wald Z | Sig.   |
|----------------------|----------|------------|--------|--------|
| Residual             | 0.0036   | 0.0004     | 9.000  | < .001 |
| Participant Variance | 0.0025   | 0.0011     | 2.272  | .023   |